Muhammad Faheem

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EXPERIENCE

National Centre for Physics (NCP) — Islamabad, Pakistan
Research Intern — July 2024 – July 2025

- Conducted a comparative analysis of classical and deep learning-based object tracking methodologies, with primary focus on single-object tracking (SOT) architectures, frameworks, and evaluation protocols in research-driven environments
- Researched deep learning-based fault detection and anomaly detection techniques for HVAC systems, leading to an accepted IEEE conference paper titled *"Deep Learning-Based Fault Detection Framework for Predictive Maintenance in HVAC Systems"* at the 13th IEEE Conference on Technologies for Sustainability (SusTech 2026)

National Aerospace Science & Technology Park (NASTP) — Rawalpindi, Pakistan
AI Engineering Intern — June 2025 – August 2025

- Designed and implemented an LLM-assisted OCR pipeline for business card digitization, enabling structured entity extraction (names, roles, organizations, contact details) and integration with an internal contact management system
- Conducted a research-focused implementation of an offline, gRPC-based multi-person behavior analysis system, integrating object detection, multi-object tracking (MOT), human pose estimation, and spatio-temporal action recognition for controlled evaluation and benchmarking

PROJECTS

Air Emissions Regional Intelligence System

- Architected a geospatial backend using PostgreSQL + PostGIS, Redis, and S3 object storage; integrated NASA Harmony OGC APIs for TEMPO multi-gas ingestion, reducing spatial query latency by **46%**
- Designed a Unified Pollution Exposure Score (UPES) and pollution-aware routing engine using OSMnx with multi-objective optimization, enabling **18% lower exposure routes** compared to shortest-path baselines while maintaining **sub-220 ms route computation latency**

Agentic RAG Multi-Project Research Lab

- Designed a secure, multi-tenant RAG framework with strict project-level isolation, PII masking, embedding-based retrieval, and evaluation harnesses (precision@k, retrieval recall, hallucination rate monitoring)
- Integrated automated red-team testing and retrieval benchmarking across **10k+ document corpora**, reducing hallucination incidence by **27%** under adversarial prompting (*Demo: github.com/username/secure-agentic-rag*)

Multi-person Behaviour Recognition System

- Implemented an offline, modular gRPC pipeline integrating YOLO-based detection, DeepSORT-style MOT, pose estimation, and spatio-temporal action recognition for **simultaneous tracking of 10+ subjects per frame**
- Optimized inter-service communication and batching, reducing pipeline latency by **31%** and enabling **sub-120 ms end-to-end processing** on multi-camera video sequences (*Demo: github.com/username/mp-behavior-recognition*)

EDUCATION

National University of Computer and Emerging Sciences — Islamabad, Pakistan
B.S. Artificial Intelligence — Aug 2022 – Present

- Honors: Deans List (Spring 2023, Fall 2025) | GPA: 3.21/4.00
- Relevant Coursework: Generative AI, Agentic AI, Reinforcement Learning, Computer Vision, ANN, Machine Learning, Digital Image Processing, NLP, Data Structures, Algorithms

SKILLS

Programming & Frameworks: Python, C/C++, JavaScript, TypeScript, FastAPI, React/Next.js
AI & ML: PyTorch, TensorFlow, Keras, scikit-learn, YOLO, ONNX, TensorRT, Hugging Face, LangChain, LangGraph, n8n
Data & Systems: NumPy, Pandas, Git, Docker, Linux, AWS, GCP, MCP